

Lucianu Dragos - Adrian
grupa 131

Examen S.A.i.

Problema 1

$$\tau = (12345)$$

$$(1) \quad \tau^2 = \tau, \quad \tau \in S_5$$

$$\text{ord}(\tau) = 5 \Rightarrow \tau^5 = e$$

$$\tau \text{ ciclu de lungime } 5 \Rightarrow \tau^5 = e$$

$$\tau^2 = \tau$$

$$\tau^6 = \tau^3 \Rightarrow \tau = \tau^3 = (12345)^3 = (14253)$$

$$(2) \quad \tau^{2021} = \tau, \quad \tau \in S_5$$

$$\tau \text{ - ciclu de lungime } 5 \Rightarrow \tau^5 = e$$

$$\tau^{2021} = \tau^1 = \tau \Rightarrow \tau = (12345)$$

$$(3) \quad \text{ord}(\tau) = 5 \Rightarrow \tau^5 = e$$

$$H = \langle \tau \rangle = \{ e, \tau, \tau^2, \tau^3, \tau^4 \}$$

$$|H| = 5$$

$$(4) \quad [S_5 : H] = \frac{|S_5|}{|H|} = \frac{5!}{5} = \frac{120}{5} = 24 \quad (\text{cf. lui Lagrange})$$

$$(5) \quad H \text{ nu este subgrup normal in } S_5$$

$$(3) \quad \tau \in S_5 \text{ a.c. } \tau H \tau^{-1} \neq H$$

$$(3) \quad \tau \in S_5 \text{ a.c. } \tau \tau \tau^{-1} \notin H$$

$\Rightarrow$ generatorul lui H

$$H = \{ e, \tau, \tau^2, \tau^3, \tau^4 \}$$

$$\tau^2 = (13524)$$

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$$\tau^3 = (14253)$$

$$\tau^4 = (15432)$$

$$H = \{ e, (12345), (13524), (14253), (15432) \}$$

$$\zeta \tau \zeta^{-1} = (\zeta(1) \zeta(2) \zeta(3) \zeta(4) \zeta(5))$$

$$\text{fie } \zeta = (45)$$

$$\zeta \tau \zeta^{-1} = (12354) \notin H \Rightarrow H \not\trianglelefteq S_5$$

$$(b) \quad H \subseteq K \trianglelefteq S_5$$

↘ minimal

$$\tau \in K \quad \left\{ \begin{array}{l} \Rightarrow \zeta \tau \zeta^{-1} \in K, \quad (b) \quad \zeta \in S_5 \\ K \trianglelefteq S_5 \end{array} \right.$$

$\Rightarrow K$ conține toți ciclul de lungime 5 (1)

fie π ciclu de lungime 5 $\Rightarrow \text{ord}(\pi) = 5$

$$\text{sgn}(\pi) = (-1)^4 = 1$$

totul ciclul de lungime 5 permutări pare (2)

A_5 - mulțimea permutărilor pare din S_5 (5)

$$A_5 \trianglelefteq S_5$$

$$\textcircled{1}, \textcircled{2}, \textcircled{5} \Rightarrow K = A_5$$

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Problema 2

$$\begin{aligned} (1) \quad x^4 - 1 &= (x^2 - 1)(x^2 + 1) \\ x^4 - 1 \in \mathbb{I} &\Leftrightarrow (\exists) u(x) \in \mathbb{K}[x] \text{ a } (x^2 - 1)u(x) = x^4 - 1 \\ (x^2 - 1)(u(x)) &= (x^2 - 1)(x^2 + 1) \Rightarrow u(x) = x^2 + 1 \in \mathbb{K}[x] \end{aligned} \quad \left. \vphantom{\begin{aligned} x^4 - 1 \in \mathbb{I} \\ (x^2 - 1)(u(x)) = (x^2 - 1)(x^2 + 1) \end{aligned}} \right\} \Rightarrow$$
$$\Rightarrow x^4 - 1 \in \mathbb{I}$$

$$x^4 - 1 \in \mathbb{J} \Leftrightarrow (\exists) v(x) \in \mathbb{K}[x] \text{ a } (x^3 - 1)v(x) = x^4 - 1$$

$$(x^3 - 1)v(x) = x^4 - 1$$

$$v(x) \text{ -> polinom de grad 1 } \Rightarrow v(x) = x - a, \quad a \in \mathbb{K}$$

$$(x^3 - 1)(x - a) = x^4 - 1$$

$$x^4 - ax^3 - x + a = x^4 - 1 \quad \text{imposibil} \Rightarrow$$

$$\cancel{(\exists)} v(x) \in \mathbb{K}[x] \text{ a } (x^3 - 1)v(x) = x^4 - 1 \Rightarrow x^4 - 1 \notin \mathbb{J}$$

$$(2) \quad \mathbb{I} = (x^2 - 1) = ((x - 1)(x + 1))$$

$$\mathbb{J} = (x^3 - 1) = ((x - 1)(x^2 + x + 1))$$

$$\begin{aligned} \mathbb{I} + \mathbb{J} &= ((x - 1)(x^2 + x + 1 + x + 1)) \\ &= ((x - 1)(x^2 + 2x + 2)) \end{aligned}$$

$$\mathbb{I} \cap \mathbb{J} = (x - 1)$$

$$(3) \quad \mathbb{K}[x] / \mathbb{I} \simeq \mathbb{K} \times \mathbb{K}$$

$$\mathbb{I} = ((x - 1)(x + 1))$$

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$$\mathbb{R}[x^2-1] \cong \mathbb{R}[x-1][x+1]$$

demonstrăm că $(x-1)$ și $(x+1)$ sunt comaximale

$$(x+1) - (x-1) = 2$$

$$\frac{1}{2}(x+1) - \frac{1}{2}(x-1) = \frac{1}{2} \cdot 2 = 1 \quad \text{comaximale } (x-1) \text{ și } x+1$$

$$\mathbb{R}[(x-1)(x+1)] \cong \mathbb{R}[x-1] \cap \mathbb{R}[x+1] \quad \textcircled{1}$$

$$\mathbb{R}[x-1] \cap \mathbb{R}[x+1] \cong \mathbb{R}[x-1] \times \mathbb{R}[x+1]$$

$\mathbb{R}[x-1]$ și să rezulte un polinom de grad 0

$$\mathbb{R}[x-1] = \{a \mid a \in \mathbb{R}\}, \text{ deci } \mathbb{R}[x-1] \cong \mathbb{R} \quad \textcircled{2}$$

$\mathbb{R}[x+1]$ și să rezulte un polinom de grad 0

$$\mathbb{R}[x+1] = \{b \mid b \in \mathbb{R}\}, \text{ deci } \mathbb{R}[x+1] \cong \mathbb{R} \quad \textcircled{3}$$

$$\textcircled{1,2,3} \Rightarrow \mathbb{R}[x^2-1] \cong \mathbb{R} \times \mathbb{R}$$

$$(4) \quad \mathbb{R}[x]/f \cong \mathbb{R} \times \mathbb{C}$$

$$f = (x-1)(x^2+x+1)$$

$$\Delta = 1-4 = -3 = (i\sqrt{3})^2$$

$$x_{1,2} = \frac{-1 \pm i\sqrt{3}}{2}$$

$$f = (x-1)\left(x - \frac{-1+i\sqrt{3}}{2}\right)\left(x - \frac{-1-i\sqrt{3}}{2}\right)$$

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(5) din (3) avem că $\mathbb{R}[X]/I = \mathbb{R} \times \mathbb{R}$
din (4) avem că $\mathbb{R}[X]/J = \mathbb{R} \times \mathbb{C}$
și cum $\mathbb{R} \times \mathbb{R} \neq \mathbb{R} \times \mathbb{C}$ $\Rightarrow \mathbb{R}[X]/I \neq \mathbb{R}[X]/J$

continuare (4)

demonstrăm că $(x-1), (x - \frac{-1+i\sqrt{3}}{2}), (x - \frac{-1-i\sqrt{3}}{2})$ sunt maximale

$$-2(x-1) + (x - \frac{-1+i\sqrt{3}}{2}) + (x - \frac{-1-i\sqrt{3}}{2}) =$$

$$-2x + 2 + 2x - 1 = 1 \text{ unitate maximală}$$

$$\mathbb{R}[X]/J = \mathbb{R}/(x-1) \times \mathbb{R}/(x - \frac{-1+i\sqrt{3}}{2}) \times \mathbb{R}/(x - \frac{-1-i\sqrt{3}}{2})$$

$$\mathbb{R}/(x-1) = \{a \mid a \in \mathbb{R}\} \text{ deci } \mathbb{R}/(x-1) \cong \mathbb{R} \quad (1)$$

$$\mathbb{R}/(x - \frac{-1+i\sqrt{3}}{2}) \times \mathbb{R}/(x - \frac{-1-i\sqrt{3}}{2}) = \mathbb{C} \quad (2)$$

$$(1)(2) \Rightarrow \mathbb{R}[X]/J = \mathbb{R} \times \mathbb{C}$$

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Problema 3

a) ireductibilitatea lui p în $\mathbb{Z}[x]$ de n , peste $\mathbb{Q}$

$$p(x) = x^3 + nx^2 - 5, \quad n \in \mathbb{Z}$$

de $\mathbb{Z}$, $(p, 2) = 1$ răd din $\mathbb{Q}$ a polinomului p , at :

$$\begin{aligned} p(1) &= 1 + n - 5 = n - 4 = 0 \Rightarrow n = 4 \\ p(-1) &= -1 + n - 5 = n - 6 = 0 \Rightarrow n = 6 \end{aligned} \quad \Rightarrow \quad \mathbb{Z} = \{\pm 1, \pm 5\}$$

$$p(1) = 1 + n - 5 = n - 4 = 0 \Rightarrow n^2 - 4 = 0 \Rightarrow n = \pm 2 \in \mathbb{Z}$$

$$p(-1) = -1 + n - 5 = n - 6 = 0 \Rightarrow n^2 - 6 = 0 \Rightarrow n = \pm \sqrt{6} \notin \mathbb{Z}$$

$$p(5) = 125 + 25n^2 - 5$$

$$= 120 + 25n^2 = 0 \Rightarrow n^2 = -\frac{120}{25} \text{ imposibil } (n^2 \geq 0)$$

$$p(-5) = -125 + 25n^2 - 5$$

$$= -130 + 25n^2 = 0 \Rightarrow n^2 = \frac{130}{25} \Rightarrow n = \pm \sqrt{\frac{130}{25}} \notin \mathbb{Z}$$

niciuna posibilitate $n = \pm 2 \Rightarrow n^2 = 4$
at a fi reductibil

$$p(x) = x^3 + 4x^2 - 5$$

$$= x^3 - x^2 + 5x^2 - 5$$

$$= x^2(x - 1) + 5(x^2 - 1)$$

$$= x^2(x - 1) + 5(x - 1)(x + 1)$$

$$= x^2(x - 1) + (5x + 5)(x - 1)$$

$$= (x^2 + 5x + 5)(x - 1)$$

$$\text{fie } q = x^2 + 5x + 5$$

$$\Delta = 25 - 20 = 5$$

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$$H_{1,2} = \frac{-5 \pm \sqrt{5}}{2} \notin \mathbb{Q}$$

i) pentru $n \neq \pm 2 \Rightarrow p(x)$ ireductibil
ii) deci $p(x) = (x^2 + 5x + 5)(x - 1)$ care in fapt ireductibili $n = \pm 2$

b) ireductibilitatea peste $\mathbb{F}_2$

$$-\hat{5} = \hat{1}$$

$$p(x) = x^3 + \hat{n}^2 x^2 + \hat{1}$$

un polinomul este de grad 3, este de asemenea este
irreductibil o ra n imparba intr-un polinom de gradul
1 si unul de grad 2.

$$p(\hat{0}) = \hat{0} + \hat{0} + \hat{1} = \hat{1} \neq \hat{0}$$

$$p(\hat{1}) = \hat{1} + \hat{n}^2 + \hat{1} = \hat{n}^2 = \hat{0} \Rightarrow \hat{n} = \hat{0}$$

$$\hat{n}^2 = \hat{0}$$

$$p(x) = x^3 + \hat{1} = \underline{x^3 - \hat{1}}$$

$$= x^3 - \hat{1}$$

$$= (x - \hat{1})(x^2 + x + \hat{1})$$

$$\text{fie } g = x^2 + x + \hat{1}$$

$$\begin{cases} g(\hat{0}) = \hat{1} \\ g(\hat{1}) = \hat{1} \end{cases} \Rightarrow g \text{ irred}$$

$\Rightarrow p(x) = (x - \hat{1})(x^2 + x + \hat{1})$ descompunerea in fact ireductibili

i) pentru $n = \hat{0} \Rightarrow p(x)$ reductibil

ii) pentru $n = \hat{1} \Rightarrow p(x)$ ireductibil

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c) ireductibilitatea peste $\mathbb{F}_3$

$$-\bar{5} = \bar{1}$$

$$p(x) = x^3 + nx^2 + x + \bar{1}$$

un polinom este de grad 3, de acesta este reducibil o sa il impartim intr-un polinom de gradul 1 si unul de grad 2

$$p(\bar{0}) = \bar{1} \neq 0$$

$$p(\bar{1}) = \bar{n}^2 + \bar{2} = \bar{0} \Rightarrow \bar{n}^2 = -\bar{2}$$

$$\bar{n}^2 = \bar{1} \Rightarrow \bar{n}^2 = \pm \bar{1} \Rightarrow n = \{\bar{1}, \bar{2}\}$$

$$p(\bar{2}) = \bar{8} + \bar{n}^2 + \bar{1} = \bar{n}^2 = \bar{0} \Rightarrow \bar{n}^2 = \bar{0} \Rightarrow \bar{n} = \bar{0}$$

i) pt $\bar{n} = \{\bar{1}, \bar{2}\} \Rightarrow \bar{n}^2 = \bar{1}$

$$p(x) = x^3 + x^2 + x + \bar{1}$$

$$= (x - \bar{1})(x^2 - x - \bar{1})$$

$$\text{fie } h = x^2 - x - \bar{1}$$

$$\left. \begin{array}{l} h(\bar{0}) = \bar{2} \\ h(\bar{1}) = \bar{2} \\ h(\bar{2}) = \bar{1} \end{array} \right\} h \text{ irred}$$

$$\begin{array}{r|l} x^3 + x^2 + \bar{1} & x - \bar{1} \\ -x^3 + x^2 & \hline \hline 2x^2 + \bar{1} & \end{array}$$

ii) pt $\bar{n} = \bar{0} \Rightarrow \bar{n}^2 = \bar{0}$

$$p(x) = x^3 + \bar{1}$$

$$= (x + \bar{1})(x^2 - x + \bar{1})$$

$$= (x + \bar{1})(x^2 + 2x + \bar{1})$$

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$$p(x) = (x + i) (x + i)^2 \\ = (x + i)^3$$

deci pt (b) $\hat{\alpha} \in \mathbb{C}$ $\Rightarrow p(x)$ reducibil